

Connect databses in Python

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Summary

There are many methods to connect db in python. We introduce two packages sqlalchemy and pycpg2. We assume .env file in which connection settings are defined.

RDB

sqlalchemy

```
import os
import sys
import sqlalchemy
from os.path import join, dirname
from dotenv import load_dotenv
from sqlalchemy.ext.declarative import declarative_base
from sqlalchemy import Column, Integer, String, create_engine
from sqlalchemy.orm import sessionmaker
```

```
import pandas as pd
import time
dotenv_path = join(dirname("$home"), '.env')
load_dotenv(dotenv_path)
conn_aurora = '{}://{}:{}_@{}:{}_/' .format('postgresql', os.environ['WRITE_RDB_USERNAME'], o
conn_redshift = '{}://{}:{}_@{}:{}_/' .format('postgresql', os.environ['DWH_USERNAME'], os.en
engine = create_engine(conn_redshift, echo=True)
```

```
def sql(query):
    session = sessionmaker(bind=engine)()
    df = pd.read_sql_query(sql=query, con=engine)
    time.sleep(1)
    session.close()
    return df
```

example query

```
q = "select id from companies limit 10"
```

```
sql(q)
```

```
2021-07-29 19:56:34,134 INFO sqlalchemy.engine.Engine select version()
2021-07-29 19:56:34,136 INFO sqlalchemy.engine.Engine [raw sql] {}
2021-07-29 19:56:34,165 INFO sqlalchemy.engine.Engine select current_schema()
2021-07-29 19:56:34,165 INFO sqlalchemy.engine.Engine [raw sql] {}
2021-07-29 19:56:34,206 INFO sqlalchemy.engine.Engine select id from companies limit 10
2021-07-29 19:56:34,208 INFO sqlalchemy.engine.Engine [raw sql] {}
```

	id
0	1
1	2
2	3
3	4
4	5
5	6
6	7
7	8
8	9

	id
9	10

psycopg2

```
import psycopg2
import pandas as pd
import time
from sshtunnel import SSHTunnelForwarder

def queryRedshift(sql):
    conn = psycopg2.connect(
        host=os.environ['DWH_HOST'],
        port=os.environ['DWH_PORT'],
        dbname=os.environ['DWH_DATABASE'],
        user=os.environ['DWH_USERNAME'],
        password=os.environ['DWH_PASSWORD'])
    cur = conn.cursor()
    cur.execute(sql)
    result = cur.fetchall()
    colnames = [col.name for col in cur.description]
    # pandas.DataFrame
    new_result = [[one for one in one_result] for one_result in result]
    result = pd.DataFrame(new_result, columns=colnames)
    cur.close()
    conn.close()
    # 1
    time.sleep(1)
    return result
```

```
queryRedshift(q)
```

	id
0	1
1	2
2	3
3	4
4	5
5	6
6	7

id	
7	8
8	9
9	10

Bigquery

It is known that the bq performance in python is depends on a connection method.
<https://medium.com/@davide.sarra/slow-bigquery-results-no-more-8aa4dde92613>

Lets compare short and long time queries ## short time query

```
query_short_time="""
select id from {}.{}.companies
""".format(os.environ['BQ_PROJECT_NAME'], os.environ['BQ_DATASET_NAME'])
```

```
start=time.perf_counter()
pd.read_gbq(query_short_time, os.environ['BQ_PROJECT_NAME'])
print(time.perf_counter()-start)
```

1.206214640999974

```
start=time.perf_counter()
pd.read_gbq(query_short_time, os.environ['BQ_PROJECT_NAME'], use_bqstorage_api=True)
print(time.perf_counter()-start)
```

2.5740085850000014

Long time query

```
query_long_time="""
omit
""";
```

```
start=time.perf_counter()
pd.read_gbq(query_long_time, os.environ['BQ_PROJECT_NAME'])
print(time.perf_counter()-start)
```

209.955542535

```
start=time.perf_counter()
pd.read_gbq(query_long_time, os.environ['BQ_PROJECT_NAME'], use_bqstorage_api=True)
print(time.perf_counter()-start)
```

10.184412958999985

cf) Long time query in Redshift

```
query="""
omit
""";
```

```
start=time.perf_counter()
sql(query)
print(time.perf_counter()-start)
```

49.52570845600002